Shady Mohamed Abdelhady

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PROFESSIONAL SUMMARY

High-ranking Mechatronics Engineering graduate (**2nd in class, Honors**) with cross-disciplinary expertise spanning **Model-Based Design (MBD), autonomous robotics, embedded systems, and software engineering**. Proven track record architecting **MIL/SIL co-simulation platforms** in MATLAB/Simulink, developing multi-domain **EV powertrain and battery models**, and deploying **perception and control stacks** in C++, Python, and ROS2. Backed by hands-on industrial experience in automotive testing, vehicle quality assurance, and automation logic design. Seeking an entry-level engineering role to build high-reliability **autonomous, robotic, or mechatronic systems**.

SKILLS

Programming Languages: C/C++, Python, SQL, MATLAB

Frameworks: ROS2, TensorFlow, PyTorch, OpenCV, CUDA

Embedded & Control: Simulink, Stateflow, UART, SPI, CAN, I2C, TCP/IP, LoRa, ESP32, Arduino, Raspberry Pi

Sensors: LiDAR, Camera, Radar, Ultrasonic, IMU, GPS

Simulation: Simscape, Gazebo, CARLA, Unity, RViz2

Tools: Git, GitHub, Jira, Ubuntu/Linux

Hardware Design: Altium Designer, KiCAD

EXPERIENCE

Process Automation Intern, Schneider Electric **07/2025**
- Analyzed industrial automation architectures and distributed control systems (DCS), mapping **functional control logic** from operational requirements to robust **ladder/FSM** implementations.
- Translated **systems engineering requirements** into actionable wiring schematics, ensuring adherence to strict **functional safety** and industrial automation **standards**.

Systems & Quality Intern, Arab American Vehicle Company **07/2024**
- Performed systematic troubleshooting, electrical diagnostics, and **root-cause failure analysis** in the after-sales center to resolve complex vehicle powertrain and chassis faults.
- Conducted rigorous quality assurance, **ECU signal checks**, and **functional safety assessments** on automotive production lines under high-volume manufacturing specifications.

Manufacturing & Testing Intern, Arab Organization for Industrialization **06/2024**
- Supported the assembly, electrical subsystem integration, and end-of-line quality verification for light electric vehicles (**electric scooters**), focusing on powertrain and battery harness reliability.
- Executed standardized **testing and validation protocols** in materials and component testing laboratories to verify operational durability, thermal limits, and mechanical integrity.

EDUCATION

Bachelor of Science in Mechatronics Engineering **09/2021 – 06/2026**
Egyptian Academy for Engineering and Advanced Technology
Graduation Project: A+, CGPA: 3.744/4, Rank: 2nd, Excellent with Honors

PROJECTS

Anti-Drift System & Digital Twin for Autonomous UAVs (Graduation Project) **09/2025 – 06/2026**
- Led a 7-member team to architect a 100 Hz MIL/SIL **co-simulation platform** bridging MATLAB/Simulink and Unity via custom TCP/IP, **calibrating the digital twin** with modeled sensor imperfections against live telemetry to close the **sim-to-real gap**.
- Executed **Grey-Box System Identification** to achieve 1**.44 cm vertical RMSE** and decouple propulsion parameters (**0.10% variance**), then verified control robustness via a $\mathbf{2^3}$ factorial stress test against compounded wind, torque, and noise disturbances.

Electric Go-Kart Powertrain & Motor Control Design **09/2026 – 09/2026**
- Modeled and integrated an electric motor subsystem into a full-vehicle multi domain simulation using Simscape incorporating **chain-drive inertia, tire friction and aerodynamic drag**.
- Sized candidate motors, tuned closed-loop **torque/current controller gains**, and benchmarked 0-target acceleration, slip ratio, and torque delivery to select the **optimal powertrain**.

Autonomous Vehicle Visual Perception Stack **08/2026 – 08/2026**
- Developed a 3D drivable space and lane boundary estimation pipeline using **semantic segmentation** masks, depth-to-3D back-projection, and **plane fitting**.
- Implemented an **obstacle validation algorithm** combining 2D bounding boxes with segmentation priors to filter false positives and compute **minimum distance to impact**.

Modular Electro-Thermal Battery Pack Design **08/2026 – 08/2026**
- Designed and sized an 18650-cell modular battery pack using **Simscape Battery Builder** to meet multi-tool power and energy constraints.
- Built coupled **electro-thermal models** to evaluate temperature rise, voltage sag, and State of Charge (SOC) dynamics across **variable discharge rates** and extreme ambient temperatures.

Simulink Doc Agent - AI Systems Engineering Copilot **05/2026 – 06/2026**
- Built an end-to-end RAG pipeline that extracts **Simulink/Stateflow model topology** (DAG signal flows, FSM states, hardware configurations) via the MATLAB Engine API and converts it into a structured, **LLM-optimized knowledge base**.
- Engineered deterministic **Markdown chunking** using Jinja2 templates to prevent fragmentation of structured data during vector embedding, improving retrieval fidelity.
- Deployed a production-grade async **FastAPI backend** (lifespan pattern, threadpool delegation) with a ChromaDB vector store (3072-d Gemini embeddings) and a Streamlit conversational interface for real-time architecture querying.

Image Processing Library in C++ **04/2026 – 04/2026**
- Built a dependency-free image processing library from scratch in pure C++11, implementing grayscale conversion (**ITU-R BT.601**), horizontal/vertical flipping, brightness & contrast adjustment, **3×3 box blur** (convolution kernel), and **90°** rotation on PPM images.
- Represented images as **3D matrices [H×W×C]** and applied both point-wise pixel operations and convolution-based filtering with border handling and value clamping.

Autonomous Unmanned Ground Vehicle **04/2025 – 07/2025**
- Developed SLAM and navigation stacks using ROS2 and LiDAR sensors for an autonomous vehicle competing in **ITC '25**.
- Configured URDF models and utilized RViz2 for real-time sensor data and coordinate transform (TF) visualization.

Motion Control of a 6DOF Robot Arm **04/2025 – 06/2025**
- Implemented multi-axis **motion planning and actuator trajectory control** using Stateflow finite state machines and an ESP32 microcontroller.
- Solved **forward and inverse kinematics** in MATLAB, generating **joint-space control trajectories** for deterministic multi-axis servo positioning.

Custom Arduino Nano PCB Design **05/2025 – 05/2025**
- Recreated and routed the official Arduino Nano hardware schematic using **Altium Designer**, executing the **end-to-end design flow** from schematic capture to custom layout and component sourcing.
- Analyzed datasheets and pinout specifications to accurately position components and route copper traces, ensuring **signal integrity** and manufacturing readiness.

Adaptive Cruise Control on RC-Scale Car **04/2025 – 05/2025**
- Designed and implemented a closed-loop longitudinal Adaptive Cruise Control system, executing **real-time** distance-based speed regulation on **embedded hardware**.
- Integrated ultrasonic distance feedback with PWM motor drive controls using a tuned **discrete PID controller**, preventing collision during sudden target vehicle deceleration.

COURSES

Visual Perception for Self-Driving Cars
UoT/Coursera

Self-Driving Specialization
Packt/Coursera

Software Engineering Diploma
Fuzetek

Model-Based Development Diploma
OS Academy/Udemy

Generative AI Bootcamp
ITI

AI & Machine Learning Diploma
Route Academy

Electric Motor Modeling & Control
MathWorks/Coursera

Mini ROS Diploma
Sector B5/ITIDA

Battery Design & Management
MathWorks/Coursera

System Programming in Linux
Eng. Reda Maher/Udemy

LANGUAGES

Arabic (Native) | **English** (Proficient)